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# Estimating Daily Reference Crop Evapotranspiration in Northeast China Using Optimized Empirical Models Based on Heuristic Intelligence Algorithms

**Zongyang Li [1,2], Zhengxin Zhao [1,2], Liwen Xing [3,*], Lu Zhao [3], Ningbo Cui [3] and Huanjie Cai [1,2,*]**

1 College of Water Resources and Architectural Engineering, Northwest A & F University, Yangling 712100, China; lzy2021050980@nwafu.edu.cn (Z.L.); 15840121398@nwafu.edu.cn (Z.Z.)
2 Institute of Water-Saving Agriculture in Arid Areas of China, Northwest A & F University, Yangling 712100, China
3 College of Water Resource and Hydropower, Sichuan University, Chengdu 610000, China; zhaolu@stu.scu.edu.cn (L.Z.); cuiningbo@stu.scu.edu.cn (N.C.)
* Correspondence: stwen028@gmail.com (L.X.); huanjiec666@163.com (H.C.)

**Abstract:** Accurately estimating reference crop evapotranspiration (ETo) improves agricultural water use efficiency. However, the accuracy of ETo estimation needs to be further improved in the Northeast region of China, the country's main grain production area. In this research, meteorological data from 30 sites in Northeast China over the past 59 years (1961–2019) were selected to evaluate the simulation accuracy of 11 ETo estimation models. By using the least square method (LSM) and three population heuristic intelligent algorithms—a genetic algorithm (GA), a particle swarm optimization algorithm (PSO), and a differential evolution algorithm (DE)—the parameters of eleven kinds of models were optimized, respectively, and the ETo estimation model suitable for northeast China was selected. The results showed that the radiation-based Jensen and Haise (JH) model had the best simulation accuracy for ETo in Northeast China among the 11 empirical models, with $R^2$ of 0.92. The Hamon model had an acceptable estimation accuracy, while the combination model had low simulation accuracy in Northeast China, with $R^2$ ranges of 0.74–0.88. After LSM optimization, the simulation accuracy of all models had been significantly improved by 0.58–12.1%. The results of heuristic intelligent algorithms showed that Hamon and Door models optimized by GA and DE algorithms had higher simulation accuracy, with $R^2$ of 0.92. Although the JH model requires more meteorological factors than the Hamon and Door model, it shows better stability. Regardless of the original empirical formula or the optimization of various algorithms, JH has higher simulation accuracy, and $R^2$ is greater than 0.91. Therefore, when only temperature or radiation factors were available, it was recommended to use the Hamon or Door model optimized by GA to estimate ETo, respectively; both models underestimated ETo with an absolute error range of 0.01–0.02 mm $d^{-1}$ compared to the reference Penman–Monteith (P–M) equation. When more meteorological factors were available, the JH model optimized by LSM or GA could be used to estimate ETo in Northeast China, with an absolute error of less than 0.01 mm $d^{-1}$. This study provided a more accurate ETo estimation method within the regional scope with incomplete meteorological data.

**Keywords:** reference crop evapotranspiration; genetic algorithm; particle swarm optimization; differential evolution; optimized empirical model

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## 1. Introduction

Evapotranspiration (ET) primarily includes two processes: water loss by soil surface evaporation and crop transpiration. ET is not only a fundamental natural process in the hydrological cycle but also an important indicator for improving water use efficiency (WUE) and optimizing irrigation scheduling. Accurate estimation of ET is crucial for effective irrigation management and enhancing WUE, particularly in hydrological and agricultural applications [1–5]. While the direct measurement of crop evapotranspiration (ET) is impractical due to the substantial expenses associated with the construction and upkeep of the necessary instruments, reference crop evapotranspiration (ETo) provides a quantitative approach to assess evaporative demand. ETo is based on a standardized reference crop—a hypothetical grassland with specific characteristics (a height of 0.12 m, a fixed surface resistance of 70 s/m, and a reflectance of 0.23) [6–8]. The Penman–Monteith (PM) method, which is recommended by the World Food and Agriculture Organization (FAO), is the standard method for determining ETo [9–12]. However, the PM needs many input meteorological factors, which may be unavailable in some regions, so its application is limited [13,14].

Over recent decades, researchers have developed numerous alternative ETo estimation models to address the limitations of the PM method. The original Penman formula was developed by Penman [15] in the United Kingdom as a combined model, with subsequent modifications adapted for different regions, including Mediterranean climates, arid regions, and temperate zones [16–21]. Temperature-based models have also been widely developed and applied, including the Hargreaves–Samani (HS) model validated in North America, the Schendel (Sch) equation developed for Central European conditions, and the Baier–Robertson (BR) model calibrated for Canadian prairie regions. Additionally, radiation-based models, such as the Priestley–Taylor (PT) model, the Makkink (Mak) model developed for Western European conditions, and the Ritchie (R) model validated in subtropical regions, have shown promising results in their respective regions. Studies comparing these models have revealed varying performance across different climatic zones [3,22,23]. For instance, Dong et al. [24] found the PT model most accurate in Northeast China, while Wu et al. [14] demonstrated better performance of the HS model for oriental arborvitae in temperate regions. These findings emphasize the importance of regional calibration and model improvements for optimal results [25–28].

Recent advances in computer science and artificial intelligence have introduced new opportunities for improving ETo estimation accuracy. Heuristic intelligent algorithms, particularly genetic algorithms (GA) and particle swarm optimization (PSO), have shown remarkable potential in agricultural and hydrological applications [29–34]. These algorithms effectively handle the complex, nonlinear relationships inherent in ETo estimation [24]. Studies have demonstrated that PSO–ELM models achieve superior accuracy in daily ETo estimation compared to traditional methods [35], while differential evolution (DE) and PSO have successfully improved Hargreaves model calibration [36]. These advancements in computational methods offer promising solutions for enhancing ETo estimation accuracy.

Northeast China, as a major grain-producing region, faces several critical challenges in ETo estimation that directly impact agricultural water management and crop production efficiency. The region suffers from insufficient meteorological station coverage, and many existing stations struggle with aging observation equipment, resulting in significant data gaps. Despite its importance in national grain production, research on ETo estimation methods specific to Northeast China remains limited, and the application of heuristic intelligent algorithms in regional water management is particularly sparse [37–39]. These challenges underscore the need for improved ETo estimation methods tailored to the region's unique characteristics and data limitations. Thus, the research objectives of this study are as follows:

(1) to reveal the spatiotemporal variation of meteorological factors and the relationship between meteorological factors and ETo in Northeast China; (2) to compare the eleven empirical models to estimate the daily ETo in Northeast China and seek the optimal ETo model in Northeast China; (3) to evaluate the effect of three heuristic intelligence algorithms on ETo estimation in Northeast China under limited meteorological input and finally obtain the ETo estimation model under the most recommended optimization algorithm.

## 2. Materials and Methods

### *2.1. Study Area and Data Collection*

The study area is the Northeast region of China, which mainly includes Liaoning, Jilin, and Heilongjiang provinces and the five eastern league cities of Inner Mongolia (38°40′–53°34′ N, 115°05′–135°02′ E) and covers an area of about $1.24 \times 10^6$ km$^2$. A digital topographic elevation map for the region is shown in Figure 1. The Northeast region of China has a temperate continental monsoon climate, which is influenced by high latitude. It has four distinct seasons, with winters being cold and lengthy, while summers are warm and brief [39]. The annual precipitation in the Northeast region of China amounts to 350–1000 mm, primarily concentrated from June to September, and the precipitation decreases from east to west in the order of humid, semi-humid, and semi-arid zones. The average temperature range is about 1.9–5.3 °C, and it can be classified into three climatic zones: the warm, temperate, and cold temperate zones. As one of the world's three major black soil areas, the Northeast region has become an important grain production base in China, occupying 12.9% of the country's land, yet accounting for 30% of the total grain production [40]. The predominant soil type here is the fertile chernozem, which supports intensive agricultural activities and contributes significantly to the region's high productivity.

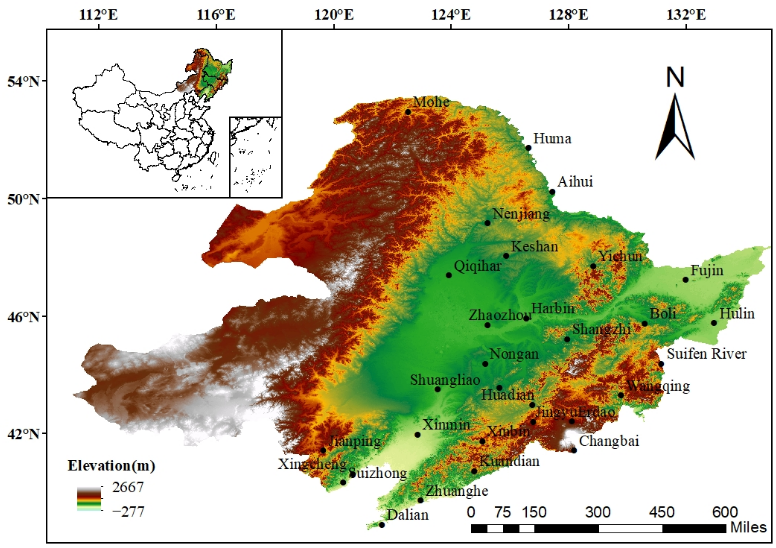


**Figure 1.** The distribution of meteorological stations in Northeast China.

In this study, 30 representative meteorological stations in Northeast China were selected, and the station data include the daily mean temperature ($T_{mean}$), maximum temperature ($T_{max}$), minimum temperature ($T_{min}$), relative humidity (RH), daily precipitation (P), wind speed at 10 m height ($u_{10}$), and sunshine duration (n) from 1961 to 2019. The data were acquired from the National Meteorological Information Center (NMIC), China Meteorological Administration (https://data.cma.cn/en, accessed on 22 September 2022). To ensure the accuracy and reliability of data, all meteorological stations are well laid out, and continuous observations were made to ensure data continuity. Additionally, all meteorological data used in this study are subject to strict quality assurance and quality control (QA/QC) procedures. This can ensure the accuracy and reliability of the data and provide a solid foundation for research [41]. Data from 1961–2007 (a total of 47 years) were utilized for model calibration to establish robust parameter estimates across a wide range of climatic conditions, while data from 2008–2019 (a total of 12 years) were reserved for independent model validation to assess performance under recent climate conditions. This division ensures both sufficient data for reliable parameter estimation and independent verification of model performance. The annual average data are shown in Table 1.

**Table 1.** Mean annual values of meteorological parameters in Northeast China during the period 1961–2019.

| Station | Lat (N) | Lon (E) | H (m) | $T_{max}$ (°C) | $T_{min}$ (°C) | $T_{mean}$ (°C) | RH (%) | $u_{10}$ (m $s^{-1}$) | $R_a$ (MJ $m^{-2}$ $d^{-1}$) | $R_n$ (MJ $m^{-2}$ $d^{-1}$) | n (h) | $ET_0$ (mm $d^{-1}$) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mohe | 52.58 | 122.31 | 438.50 | 4.59 | −11.57 | −4.04 | 0.69 | 1.40 | 23.17 | 5.62 | 6.60 | 1.55 |
| Huma | 51.44 | 126.38 | 173.90 | 5.96 | −6.96 | −0.84 | 0.66 | 1.57 | 23.68 | 5.87 | 6.98 | 1.80 |
| Aihui | 50.15 | 127.27 | 166.40 | 6.71 | −4.82 | 0.64 | 0.66 | 2.30 | 24.29 | 6.05 | 7.21 | 1.95 |
| Nenjiang | 49.10 | 125.14 | 242.20 | 7.30 | −5.79 | 0.64 | 0.66 | 2.35 | 24.73 | 6.32 | 7.28 | 2.09 |
| Keshan | 48.03 | 125.53 | 236.00 | 7.82 | −3.39 | 2.03 | 0.65 | 2.07 | 25.18 | 6.45 | 7.24 | 2.11 |
| Qiqihar | 47.23 | 123.55 | 147.10 | 10.01 | −1.35 | 4.11 | 0.60 | 2.39 | 25.45 | 6.64 | 7.73 | 2.43 |
| Yichun | 47.42 | 128.50 | 264.80 | 8.42 | −5.02 | 1.34 | 0.69 | 1.57 | 25.32 | 6.37 | 6.41 | 1.83 |
| Fujin | 47.14 | 131.59 | 66.40 | 8.56 | −1.81 | 3.18 | 0.68 | 2.75 | 25.51 | 6.39 | 6.67 | 2.06 |
| Zhaozhou | 45.42 | 125.15 | 148.70 | 10.60 | −1.25 | 4.43 | 0.63 | 2.76 | 26.11 | 7.00 | 7.88 | 2.50 |
| Harbin | 45.56 | 126.34 | 118.30 | 10.24 | −0.92 | 4.54 | 0.65 | 2.47 | 26.02 | 6.74 | 6.72 | 2.32 |
| Shangzhi | 45.13 | 127.58 | 189.70 | 9.88 | −2.80 | 3.23 | 0.72 | 2.09 | 26.30 | 6.98 | 6.73 | 2.00 |
| Boli | 45.45 | 130.36 | 234.40 | 9.90 | −0.52 | 4.63 | 0.61 | 2.56 | 26.09 | 6.54 | 6.54 | 2.34 |
| Hulin | 45.46 | 132.58 | 100.20 | 8.93 | −1.27 | 3.62 | 0.70 | 2.21 | 26.08 | 6.66 | 6.59 | 1.95 |
| Nongan | 44.23 | 125.09 | 170.20 | 11.29 | −0.45 | 5.21 | 0.63 | 2.62 | 26.62 | 7.05 | 7.17 | 2.45 |
| Suifen River | 44.23 | 131.10 | 567.80 | 8.88 | −2.30 | 3.02 | 0.67 | 2.42 | 26.62 | 6.74 | 6.59 | 1.99 |
| Shuangliao | 43.30 | 123.32 | 114.90 | 12.89 | 0.48 | 6.36 | 0.62 | 2.44 | 26.96 | 7.39 | 7.87 | 2.57 |
| Shuangyang | 43.33 | 125.38 | 219.50 | 11.67 | −0.08 | 5.56 | 0.66 | 2.16 | 26.94 | 7.09 | 6.77 | 2.31 |
| Wangqing | 43.18 | 129.47 | 244.80 | 11.63 | −1.57 | 4.46 | 0.67 | 1.41 | 27.03 | 6.86 | 6.33 | 1.96 |
| Huadian | 42.59 | 126.45 | 263.30 | 11.56 | −1.23 | 4.68 | 0.71 | 1.55 | 27.15 | 7.10 | 6.12 | 2.03 |
| Jingyu | 42.24 | 126.48 | 570.00 | 10.80 | −3.18 | 3.54 | 0.71 | 1.47 | 27.38 | 7.19 | 6.42 | 1.94 |
| Erdao | 42.24 | 128.07 | 731.7 | 10.58 | −3.73 | 3.07 | 0.71 | 1.85 | 27.37 | 7.23 | 6.52 | 1.98 |
| Jianping | 41.25 | 119.37 | 46.22 | 15.33 | 2.76 | 8.64 | 0.52 | 1.49 | 27.74 | 7.43 | 7.66 | 2.67 |
| Xinmin | 41.58 | 122.51 | 30.90 | 14.12 | 3.10 | 8.26 | 0.62 | 2.57 | 27.53 | 7.44 | 7.24 | 2.64 |
| Xinbin | 41.44 | 125.03 | 328.40 | 12.98 | −0.93 | 5.34 | 0.71 | 1.03 | 27.62 | 7.18 | 6.13 | 1.96 |
| Changbai | 41.25 | 128.11 | 775.00 | 9.60 | −3.11 | 2.68 | 0.69 | 1.75 | 27.74 | 7.28 | 6.70 | 1.97 |
| Suizhong | 40.20 | 120.18 | 29.00 | 15.57 | 4.76 | 9.76 | 0.63 | 2.03 | 28.13 | 7.64 | 7.28 | 2.58 |
| Xingcheng | 40.37 | 120.39 | 20.50 | 14.80 | 4.39 | 9.33 | 0.64 | 2.04 | 28.03 | 7.71 | 7.58 | 2.52 |
| Kuandian | 40.43 | 124.47 | 260.10 | 13.37 | 1.78 | 7.23 | 0.70 | 1.09 | 27.99 | 7.40 | 6.51 | 2.04 |
| Zhuanghe | 39.43 | 122.57 | 34.80 | 14.30 | 4.84 | 9.25 | 0.69 | 2.11 | 28.35 | 7.61 | 6.85 | 2.33 |
| Dalian | 38.55 | 121.38 | 91.50 | 14.71 | 8.01 | 11.05 | 0.65 | 3.29 | 28.64 | 8.01 | 7.37 | 2.82 |

### *2.2. Reference Evapotranspiration Model*

#### 2.2.1. Penman–Monteith Equation

In previous studies, the PM equation was mostly used to estimate daily scale ETo [6,7,42], so this study chose to use the PM equation as a reference index for model evaluation, and the specific calculation formula is as follows:

$$ETo = \frac{0.408\Delta(R_n - G) + \gamma\frac{900}{T+273}u_2(e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)} \quad (1)$$

where ETo represents reference crop evapotranspiration rate (mm $d^{-1}$); $R_n$ represents net radiation on crop surface (MJ $m^{-2}$ $day^{-1}$), which can be calculated from the sunshine duration n; G represents soil heat flux (MJ $m^{-2}$ $day^{-1}$); T represents daily mean temperature, denoted in this study by $T_{mean}$; $u_2$ represents wind speed at height of 2 m (m $s^{-1}$); $e_s$ represents saturation water vapor pressure (kPa); $e_a$ represents actual water vapor pressure (kPa); ($e_s$–$e_a$) represents vapor pressure deficit (kPa); Δ represents the inclination rate of saturated water vapor pressure curve (kPa °C$^{-1}$); and $\gamma$ represents psychrometer constant (kPa °C$^{-1}$).

The Rn was calculated by the following equation [6]:

$$R_n = R_{ns} - R_{nl} \tag{2}$$

$$R_{ns} = (1-\alpha)R_s \tag{3}$$

$$R_{nll} = \sigma\frac{T_{max,k}^4 + T_{min,k}^4}{4}(0.34 - 0.14\sqrt{e_a})\left(1.35\frac{R_s}{R_{so}} - 0.35\right) \tag{4}$$

$$R_s = \left(a + b\frac{n}{N}\right)R_a \tag{5}$$

$$\mathrm{R_{so}} = \left(0.75 + 210^{-5}\mathrm{H}\right)\mathrm{R_a} \tag{6}$$

where $\alpha$ is the albedo or canopy reflection coefficient, which is 0.23 for hypothetical reference grass; $\sigma$ is Stefan–Boltzmann constant (4.903 × 10$^{-9}$ MJK$^{-4}$ m$^{-2}$ day$^{-1}$); $T_{max,k}$ and $T_{min,k}$ are the maximum and minimum absolute air temperatures (K = °C + 273.6); n is the actual sunshine duration (h); H is the elevation of the station (m); Ra and N are the extraterrestrial radiation (MJ m$^{-2}$ day$^{-1}$) and the maximum possible duration of sunshine (h), which are calculated by Equations (21) and (34) in FAO paper 56, respectively; and a and b are empirical regression coefficients.

In the calculation of ETo, the observed wind speed at 2 m above the surface is required. In order to convert the measured wind speed at non-2 m to the wind speed at 2 m, the logarithmic wind profile relationship is usually adopted to calculate the wind speed above the coverage of short grass. The calculation formula is as follows:

$$u_2 = u_z\frac{4.87}{\ln(67.8u_z - 5.42)} \tag{7}$$

where $u_z$ is the wind speed Z m above the surface (m s$^{-1}$), Z is the height of the measured wind speed above the surface (m), and the height in this paper is 10 m.

### 2.2.2. Temperature-Based Models

Hargreaves and Samani [43] devised a temperature-based ETo estimation model constructed by incorporating the effect of solar radiation. It only needs the air temperature and solar radiation to estimate ETo, as shown in the following formula:

$$ETo = a_{HS}\frac{R_a}{\lambda}(T_{mean} + 18.5)(T_{max} - T_{min})^{b_{HS}} \tag{8}$$

where $R_a$ is extraterrestrial radiation (MJ m$^{-2}$ day$^{-1}$); $\lambda$ is the latent heat of evaporation (MJ kg$^{-1}$); $T_{mean}$ is mean air temperature (°C); $T_{max}$ and $T_{min}$ are the maximum and minimum air temperature (°C); and $a_{HS}$ and $b_{HS}$ are empirical parameters.

Baier and Robertson [44] constructed a function of temperature and $R_a$ to estimate ETo, as shown in the following formula:

$$ETo = a_{BR}T_{max} + b_{BR}(T_{max} - T_{min}) + c_{BR}R_a + d_{BR} \tag{9}$$

where $a_{BR}$, $b_{BR}$, $c_{BR}$, and $d_{BR}$ are empirical parameters.

Schendel [45] constructed the following model to estimate ETo based on the ETo and $T_{mean}$/RH ratio, as shown in the following formula:

$$ETo = a_{Sch}\frac{T_{mean}}{RH} \tag{10}$$

where RH is relative humidity (%) and $a_{Sch}$ is an empirical parameter.

Hamon [46] developed a model for estimating ETo by utilizing the adjusted average temperature squared with the day length factor, as shown in the following formula:

$$ETo = a_{Hamon}n^2 \times 4.95 \times 2.73^{b_{Hamon}T_{mean}} \tag{11}$$

where n is sunshine duration (h) and $a_{Hamon}$ and $b_{Hamon}$ are the empirical parameters.

2.2.3. Radiation-Based Models

Priestley and Taylor [47] posited that the large-scale parameterization of surface sensible and latent heat fluxes is appropriate for energy considerations over land and proposed a widely applicable equation, as shown in the following formula:

$$ETo = \frac{a_{PT}}{\lambda} \times \frac{\Delta}{\lambda + \gamma} \times (R_n - G) \tag{12}$$

where $a_{PT}$ is an empirical parameter.

Makkink [48] proposed a model that estimates ETo using solar radiation, as shown in the following formula:

$$ETo = \frac{a_{Mak}}{\lambda} \times \frac{\Delta}{\lambda + \gamma} \times (R_s - b_{Mak}) \tag{13}$$

where $R_s$ is solar shortwave irradiance (MJ m$^{-2}$ day$^{-1}$) and $a_{Mak}$ and $b_{Mak}$ are the empirical parameters.

Based on energy balance, Jensen and Haise [49] developed a model that requires only a few fundamental parameters such as $T_{mean}$, $R_a$, and n to estimate the ETo, as shown in the following formula:

$$ETo = a_{JH} \times (b_{JH}T_{mean} + c_{JH}) \times 0.408 \times (d_{JH} + \frac{d_{JH} \times n}{N}) \times R_a \tag{14}$$

where $a_{JH}$, $b_{JH}$, $c_{JH}$, $d_{JH}$, and $e_{JH}$ are empirical parameters and N is the maximum possible sunshine hours (h), which can be calculated using the following formula:

$$N = \frac{24}{\pi}\omega_s \tag{15}$$

where $\omega_s$ is the angle at sunset (rad).

The Doorenbos and Pruitt [50] version of the Penman equation was developed based on the information in Appendix 2 of the FAO 24, as shown in the following formula:

$$ETo = a_{Door} \times \frac{\Delta \times R_s}{\lambda + \gamma} + b_{Door} \tag{16}$$

where $a_{Door}$ and $b_{Door}$ are the empirical parameters.

2.2.4. Combination Models

Penman [15] proposed a composite model that has been used worldwide that can accurately estimate ETo with only a few conventional meteorological data, as shown in the following formula:

$$ETo = \frac{\Delta}{\lambda + \gamma} \times (R_n - G) + a_{48PM} \times \frac{\gamma}{\lambda + \gamma} \times (1 + b_{48PM}u_2)(e_s - e_a) \tag{17}$$

where $a_{48PM}$ and $b_{48PM}$ are the empirical parameters.

Van Bavel [20] improved the Penman model by incorporating approximate expressions for water vapor and sensible heat transport. The Penman model was improved to model the estimation of ETo for $R_n$, ambient air characteristics, and roughness of surface, as shown in the following formula:

$$ETo = \frac{(a_{PVB}T_{mean}^2 - b_{PVB}T_{mean} + c_{PVB}) \times \frac{R_n}{\lambda} + \frac{e_s - e_a}{80.8/(u_2+0.1)}}{a_{PVB}T_{mean}^2 - b_{PVB}T_{mean} + c_{PVB} + 1} \quad (18)$$

where $a_{PVB}$, $b_{PVB}$, and $c_{PVB}$ are empirical parameters.

Brutsaert and Stricker [51] believed that regional dryness can be obtained from the regional advection of the air-drying capacity implied by the atmospheric conditions, so the ETo calculation formula is as follows:

$$ETo = (2a_{BS} - 1) \times \frac{\Delta}{\lambda + \gamma} \times (R_n - G) - \frac{\lambda}{\Delta + \lambda} \times [b_{BS} \times (1 + c_{BS} \times u_2)] \times (e_s - e_a) \quad (19)$$

where $a_{BS}$, $b_{BS}$, and $c_{BS}$ are the empirical parameters.

In this study, four temperature-based models, four radiation-based models, and three mixed models were selected, and there were differences in theory and required input meteorological elements among the different models. The specific formulas and empirical parameters of the models are shown in Table 2.

**Table 2.** Estimated ETo formulas and empirical parameters based on temperature, radiation, and combination models.

| Types | Methods | Equation | Parameter | Reference |
|---|---|---|---|---|
| Temperature-based | Hargreaves–Samani | $ETo = a_{HS}\frac{R_a}{\lambda}(T_{mean}+18.5)(T_{max} - T_{min})^{b_{HS}}$ | $a_{HS}$, $b_{HS}$ | [43] |
| | Baier–Robertson | $ETo = a_{BR}T_{max} + b_{BR}(T_{max} - T_{min}) + c_{BR}R_a + d_{BR}$ | $a_{BR}$, $b_{BR}$, $c_{BR}$, $d_{BR}$ | [44] |
| | Schendel | $ETo = a_{Sch}\frac{T_{mean}}{RH}$ | $a_{Sch}$ | [45] |
| | Hamon | $ETo = a_{Hamon}n^2 \times 4.95 \times 2.73^{b_{Hamon}T_{mean}}$ | $a_{Hamon}$, $b_{Hamon}$ | [46] |
| Radiation-based | Priestley–Taylor | $ETo = \frac{a_{PT}}{\lambda} \times \frac{\Delta}{\lambda+\gamma} \times (R_n - G)$ | $a_{PT}$ | [47] |
| | Makkink | $ETo = \frac{a_{Mak}}{\lambda} \times \frac{\Delta}{\lambda+\gamma} \times (R_s - b_{Mak})$ | $a_{Mak}$, $b_{Mak}$ | [48] |
| | Jensen and Haise | $ETo = a_{JH} \times (b_{JH}T_{mean} + c_{JH}) \times 0.408 \times (d_{JH} + \frac{e_{JH} \times n}{N}) \times R_a$ | $a_{JH}$, $b_{JH}$, $c_{JH}$, $d_{JH}$ | [49] |
| | Doorenbos and Pruitt | $ETo = a_{Door} \times \frac{\Delta \times R_s}{\lambda+\gamma} + b_{Door}$ | $a_{Door}$, $b_{Door}$ | [50] |
| Combination | Penman | $ETo = \frac{\Delta}{\lambda+\gamma} \times (R_n - G) + a_{48PM} \times \frac{\gamma}{\lambda+\gamma} \times (1 + b_{48PM}u_2)(e_s - e_a)$ | $a_{48PM}$, $b_{48PM}$ | [15] |
| | Van Bavel | $ETo = \frac{(a_{PVB}T_{mean}^2 - b_{PVB}T_{mean} + c_{PVB}) \times \frac{R_n}{\lambda} + \frac{e_s - e_a}{80.8/(u_2+0.1)}}{a_{PVB}T_{mean}^2 - b_{PVB}T_{mean} + c_{PVB} + 1}$ | $a_{PVB}$, $b_{PVB}$, $c_{PVB}$ | [20] |
| | Brutsaert and Stricker | $ETo = (2a_{BS} - 1) \times \frac{\Delta}{\lambda+\gamma} \times (R_n - G) - \frac{\lambda}{\Delta+\lambda} \times [b_{BS} \times (1 + c_{BS} \times u_2)] \times (e_s - e_a)$ | $a_{BS}$, $b_{BS}$ | [51] |

### *2.3. Heuristic Intelligence Optimistic Algorithm*

To comprehensively evaluate the applicability of heuristic intelligence algorithms on the ETo estimation model, the empirical parameters of the ETo model were calibrated by GA, PSO, and DE and compared with the least square method (LSM). The flow chart is shown in Figure 2.

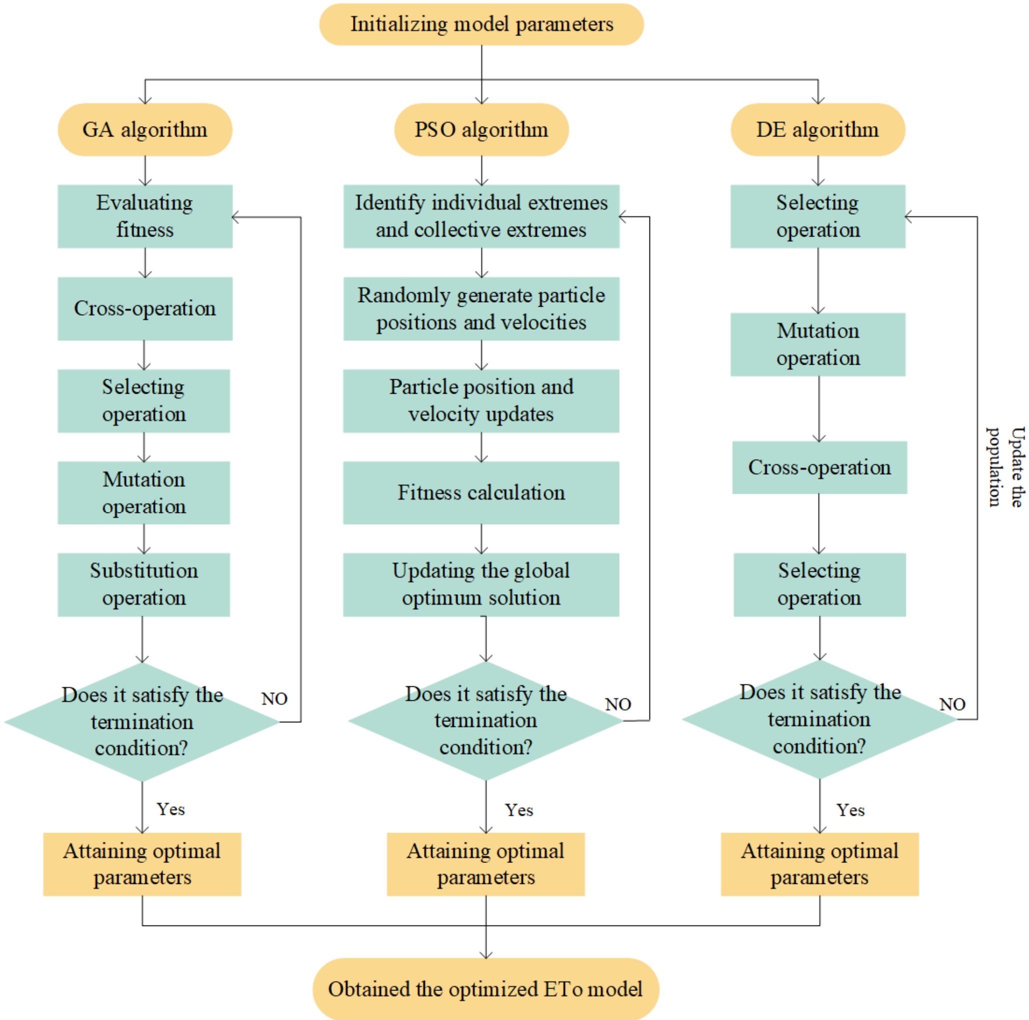


**Figure 2.** Flow structure diagram of genetic algorithm (GA), particle swarm optimization (PSO), and differential evolution (DE).

### 2.3.1. Genetic Algorithm

The GA algorithm developed by Holland [52] is a search algorithm inspired by the theory of natural evolution. It aims to search for the optimal solutions to problems by simulating genetic mechanisms, which mimic the processes of natural selection and reproduction. The GA has a wide range of applicability and is capable of finding global optimal solutions, thus providing optimal solutions for a variety of problems involving searches and optimization. Similar to the process of natural evolution, it can improve some of the disadvantages of traditional algorithms, especially for problems with many parameters and complex calculation formulas. The GA assumes that the optimal solution to the problem at hand consists of multiple elements and that when more such elements are combined, the conclusions obtained will be closer to the optimal solution of the problem. The population comprises individuals who possess the necessary components for the best possible solution. As the selection and crossover process is repeated, these components are transmitted to the next generation and may be merged with other crucial elements of the optimal solution. This results in genetic pressure that drives more individuals in the population to encompass the elements essential for the optimal solution.

After initializing the population, the algorithm uses roulette wheel selection or tournament selection to choose individuals for the next generation based on their fitness values. Roulette wheel selection probability can be seen in the following equation:

$$p_i = \frac{f(x_i)}{\sum_{j=1}^{N} f(x_j)} \tag{20}$$

where $f(x_i)$ is the fitness value of individual $x_i$.

Two parent individuals, $x_i$ and $x_j$, undergo crossover to generate two offspring individuals, $y_i$ and $y_j$. The following is a single-point crossover formula:

$$y_i = \left(x_i^1, x_i^2, \ldots, x_i^k, x_j^{k+1}, \ldots, x_j^n\right) y_j = \left(x_j^1, x_j^2, \ldots, x_j^k, x_i^{k+1}, \ldots, x_i^n\right) \tag{21}$$

where k is a randomly chosen crossover point.

The following equation mutates an individual $x_i$ at a certain position:

$$x_i' = \begin{cases} x_i + \delta, \ with\ probability\ p_m \\ x_i,\ otherwise \end{cases} \tag{22}$$

where δ is a random perturbation, and $p_m$ is the mutation probability.

### 2.3.2. Particle Swarm Optimization

The PSO algorithm is a heuristic search algorithm that simulates the collective behavior of biological swarms. The simplified model was initially proposed by James Kennedy and Russell Eberhart [53] and seeks the optimal solution by simulating the motion and information exchange of particles in the search space. The PSO algorithm performs an optimal search through the collaboration of multiple particles, and it searches by adjusting the position and speed of each particle. Through learning and exchanging information, each particle updates its position and velocity, striving to find superior solutions within the search space [35,54]. In the PSO algorithm, the update of the speed and position of each particle is calculated using the two equations below.

The speed update formula is as follows:

$$v_i(t+1) = \omega v_i(t) + c_1 r_1 (P_i - X_i) + c_2 r_2 (P_g - X_i) \tag{23}$$

The position update formula is as follows:

$$X_i(t+1) = X_i(t) + v_i(t+1) \tag{24}$$

where $v_i(t)$ denotes the velocity of particle i at the t-th iteration, $X_i(t)$ denotes the position of particle i at the t-th iteration, $P_i$ denotes the individual optimal position of particle i, $P_g$ denotes the global optimal position of all particles, ω stands for the inertia weight, $c_1$ and $c_2$ are the learning factors, and $r_1$ and $r_2$ represent empirical parameters. The PSO algorithm can iteratively update the velocity and position of the particles until a set of termination conditions is reached. Ultimately, the particles will converge around some superior solutions, thus leading to the discovery of the optimal solution to the problem.

### 2.3.3. Differential Evolution Algorithm

The DE algorithm is a heuristic search algorithm for global optimization problems. It simulates the processes of mutation, crossover, and selection in natural evolution to find the optimal solution. DE mutates individuals within the population to generate a new set of solution vectors. Subsequently, it performs a crossover between the new solution vectors and the original solution vectors, creating a further set of solution vectors. Finally, it updates the population by selecting the better-adapted individuals from the newly generated solution vectors and the original solution vectors through a selection operation [3,36,55].

For each target individual, $x_i^g$, a mutant vector, $v_i^g$, can be generated using the following equation:

$$v_i^g = x_r^g + F \times \left(x_s^g - x_t^g\right) \quad (25)$$

where r, s, and t are randomly selected different individuals from the population, with $r \neq i$, $s \neq i$, $t \neq i$, and F is the scaling factor.

Crossover between the target individual, $x_i^g$, and the mutant vector, $v_i^g$, to generate a trial individual, $u_i^g$, can be performed using the following equation:

$$u_i^g(j) = \begin{cases} v_i^g(j),\ if\ rand\ (0,1) \leq CR\ or\ j = j_{rand} \\ x_i^g(j),\ otherwise \end{cases} \quad (26)$$

where CR is the crossover probability, and $j_{rand}$ is a randomly chosen dimension index.

If the trial individual, $u_i^g$, has better fitness than the target individual, $x_i^g$, $x_i^g$ can be replaced with $u_i^g$, as follows:

$$x_i^{g+1} = \begin{cases} u_i^g,\ if\ f\left(u_i^g\right) < f\left(x_i^g\right) \\ x_i^g,\ otherwise \end{cases} \quad (27)$$

### *2.4. Model Implementation and Evaluation*

Mohammad Hossein Kazemi [56] has shown that chronological data sets can be divided into training sets and test sets using the common two-piece partitioning strategy to obtain the most accurate results. Therefore, this study used the data from 30 meteorological stations in Northeast China from 1961–2019 and divided them into two parts according to time, as follows: (1) data from 1961–2008 were used as a training period to calibrate the empirical parameters of each selected ETo model; (2) data from 2009–2019 were used as a validation period to evaluate and calibrate the accuracy of the ETo model.

Due to the lack of empirical data, following Martí et al. [57], we used the ETo values calculated by the PM equation as a reference. Therefore, we used the PM equation to calculate the standard ETo values and conducted a comprehensive evaluation of the ETo values obtained by different models using five evaluation criteria: determination coefficient ($R^2$), root mean square error (RMSE), relative root mean square error (RRMSE), Nash–Sutcliffe efficiency coefficient (NSE), and the mean absolute error (MAE) to evaluate the accuracy of different simulation results.

$$R^2 = \frac{\left[\sum_{i=1}^{m}(X_i - \overline{X})(Y_i - \overline{Y})\right]^2}{\sum_{i=1}^{m}(X_i - \overline{X})^2 \sum_{i=1}^{m}(Y_i - \overline{Y})^2} \quad (28)$$

$$RMSE = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(Y_i - X_i)^2} \quad (29)$$

$$RRMSE = \frac{\sqrt{\frac{1}{m}\sum_{i=1}^{m}(Y_i - X_i)^2}}{\overline{X}} \times 100\% \quad (30)$$

$$NSE = 1 - \frac{\sum_{i=1}^{m}(Y_i - X_i)^2}{\sum_{i=1}^{m}(X_i - \overline{X})^2} \quad (31)$$

$$\mathrm{MAE} = \frac{1}{m}\sum_{i=1}^{m}|Y_i - X_i| \quad (32)$$

where $X_i$ and $Y_i$ denote the calculated values of the i-th and the PM equation, respectively. $\overline{X}$ and $\overline{Y}$ are the mean values of $X_i$ and $Y_i$; m is the number of data points.

In this study, each indicator provides unique insights into different aspects of the study, and using these indicators allows us to understand the results more thoroughly. According to previous studies, the combination of multiple evaluation indicators can comprehensively evaluate research results, enhance the overall evaluation ability, and contribute to more detailed analysis. Therefore, we believe that it is necessary to use these evaluation indicators in our research.

### *2.5. Mann–Kendall Trend Analysis*

To evaluate the trends in $T_{mean}$, RH, n, and ETo from 1961 to 2019, we employed the Mann–Kendall (MK) trend test, a non-parametric statistical method widely used for detecting monotonic trends in time series data [58].

## 3. Results

### *3.1. Spatiotemporal Variation and Correlation Analysis of Meteorological Characteristics*

The spatial distribution and temporal variation of annual mean $T_{mean}$, RH, n, and ETo in Northeast China during the period 1961–2019 are shown in Figure 3. From 1961 to 2019, the annual average $T_{mean}$ and RH showed a significant increase and decrease trend, respectively. The spatial distribution of the annual average $T_{mean}$ decreased from west to east and from south to north, with a variation range from −4.03 to 11.00 °C. The spatial distribution of RH increased from west to east and from south to north, with a variation range of 59–70%. The average annual sunshine duration in Northeast China decreased from west to east and from south to north, with a variation range of 2235.5–2877.1 h, among which 12 sites showed a significant decreasing trend, and only one site located on the easternmost side showed a significant increasing trend. The spatial distribution of ETo calculated by using FAO56 PM decreased from west to east and from south to north, with a variation range of 567.8–1030.5 mm, with five sites showing a significant decreasing trend and six sites showing a significant increasing trend. In the eastern and northern parts of Northeast China, ETo showed a significant upward trend during the years under study, while in the central and southern parts, it showed a significant downward trend, except for Zhuanghe station. Correlation analysis of $T_{mean}$, RH, n, and ETo in Northeast China showed that ETo in Northeast China was significantly correlated with $T_{mean}$, RH, and n, with correlation coefficients of 0.82, −0.71, and 0.70, respectively. RH was negatively correlated with n and $T_{mean}$, with correlation coefficients of −0.74 and −0.36 (Figure 4). Therefore, the calculation of ETo is closely related to temperature, radiation, and humidity, and it is particularly necessary to use models with different input factors to simulate ETo reasonably.

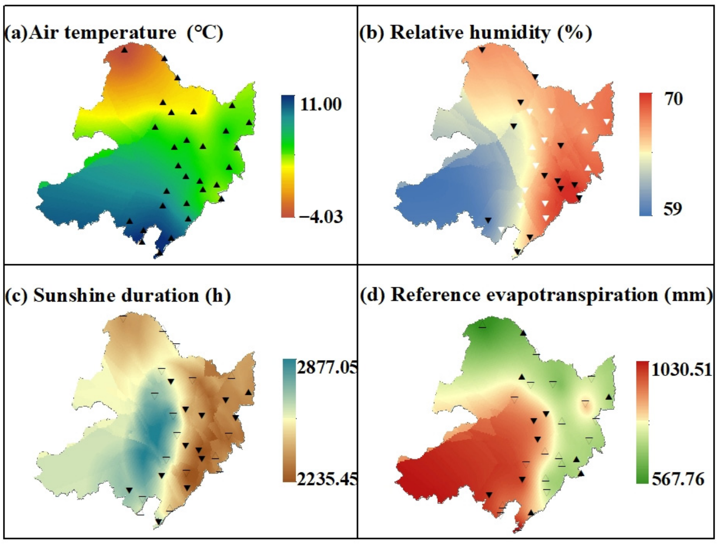


**Figure 3.** Spatial distribution and temporal trends of annual mean $T_{mean}$ (°C), RH (%), n (h), and $ET_0$ (mm/d) in Northeast China from 1961 to 2019. Notes: ▲ indicates a significant increase ($p < 0.05$), △ indicates an increase, ▼ indicates a significant decrease ($p < 0.05$), and ▽ indicates a decrease.

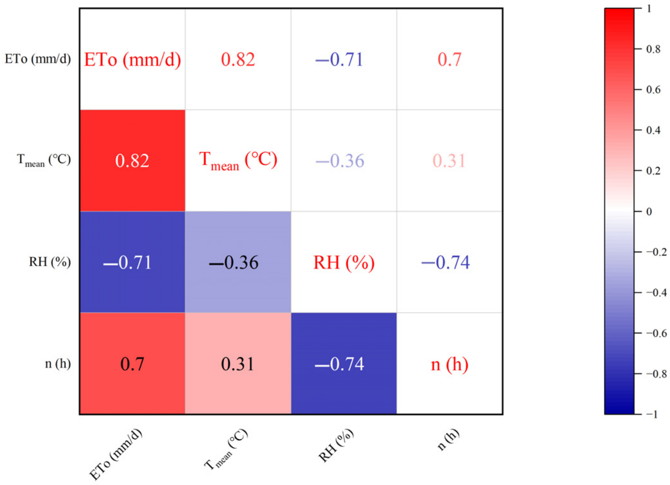


**Figure 4.** Correlation map between environmental factors in Northeast China.

### *3.2. The Simulation Accuracy of Temperature-Based Models*

Table 3 shows that the HS model had the highest calculation accuracy compared to the other three temperature-based models, and the medians of $R^2$, RMSE, RRMSE, NSE, and MAE were superior to BR and Sch models by 6.64%, 64.28%, 66.35%, 242.43%, and 67.07% and 8.10%, 18.84%, 22.24%, 8.05%, and 12.52%, respectively. The medians of $R^2$ were just lower than that of the Hamon model, but the medians of RMSE, RRMSE, NSE, and MAE were higher by 43.02%, 46.19%, 387.50%, and 52.02%, respectively.

Figure 5 shows the medians of $R^2$, RMSE, RRMSE, NSE, and MAE for four algorithms: LSM, GA, PSO, and DE. From the combination of Table 3 and Figure 5, it can be inferred that for the HS model, the accuracy of the original model is only just lower than the LSM but superior to GA, PSO, and DE. The medians of $R^2$, RMSE, RRMSE, NSE, and MAE of the LSM were better than the original model by 0.58%, 12.37%, 12.37%, 4.15%, and 3.34%, respectively. This indicated that LSM has significantly promoted the computational accuracy of the original model, but the remaining algorithms do not perform as accurately as the original model. For the HS model, the GA, PSO, and DE do not improve the model accuracy. For the BR, Sch, and Hamon models, the computational accuracy of the original models was the poorest, lower than the model accuracy computed by any of the algorithms. For the BR model, the $R^2$, RMSE, RRMSE, NSE, and MAE of LSM were improved by 2.10%, 66.44%, 67.04%, 242.43%, and 71.68% compared to the original model, which showed that the LSM has mostly improved the original model's accuracy. In the computation of the BR model, the GA, PSO, and DE consistently maintain the same level of computational accuracy, with the median values of the five evaluation indexes being identical. Moreover, the difference in $R^2$, RMSE, RRMSE, NSE, and MAE between LSM and GA, PSO, and DE was within 0.001, 0.005, 0.002, 0.001, and 0.001, respectively. The gaps in each evaluation index were not more than 0.005, which indicated that the four algorithms can significantly enhance the model accuracy for the BR model. Although optimization algorithms only slightly compared with LSM, they also have a larger improvement in the computational accuracy of the original model. Therefore, GA, PSO, and DE are all suitable for the computation of the BR model. For the Sch model, the median values of $R^2$, RMSE, RRMSE, NSE, and MAE of LSM were improved by 3.11%, 22.54%, 24.05%, 6.99%, and 18.92%, respectively, compared to the original model, indicating a significant improvement in LSM compared to the original model. In the Sch model, both GA and PSO exhibited consistent model computation accuracy, with identical values for $R^2$, RRMSE, NSE, and MAE. Only the RMSE value differed, by 0.001. Additionally, the model accuracy of these two optimization algorithms was the same as LSM in terms of $R^2$, with RMSE increasing by 0.61%, RRMSE increasing by 0.79%, NSE increasing by 0.12%, and MAE decreasing by 0.34% compared to LSM.

**Table 3.** The median values of the five evaluation metrics for temperature-based models.

| **Model** | **Model Accuracy Evaluation Metrics** | | | | |
|---|---|---|---|---|---|
| | **$R^2$** | **RMSE** | **RRMSE** | **NSE** | **MAE** |
| HS | 0.87 | 0.86 | 0.39 | 0.82 | 0.63 |
| LSM–HS | 0.87 | 0.75 | 0.34 | 0.85 | 0.61 |
| GA–HS | 0.55 | 1.50 | 0.68 | 0.47 | 1.25 |
| PSO–HS | 0.07 | 3.06 | 1.30 | −0.07 | 2.31 |
| DE–HS | 0.01 | 2.60 | 1.27 | −0.48 | 2.06 |
| BR | 0.81 | 2.40 | 1.15 | −0.58 | 1.91 |
| LSM–BR | 0.83 | 0.81 | 0.38 | 0.82 | 0.54 |
| GA–BR | 0.83 | 0.80 | 0.38 | 0.82 | 0.54 |
| PSO–BR | 0.83 | 0.80 | 0.38 | 0.82 | 0.54 |
| DE–BR | 0.83 | 0.80 | 0.38 | 0.82 | 0.54 |
| Sch | 0.80 | 1.06 | 0.50 | 0.76 | 0.72 |
| LSM–Sch | 0.83 | 0.82 | 0.38 | 0.81 | 0.58 |
| GA–Sch | 0.83 | 0.81 | 0.38 | 0.81 | 0.59 |
| PSO–Sch | 0.83 | 0.81 | 0.38 | 0.81 | 0.59 |
| DE–Sch | 0.81 | 0.82 | 0.38 | 0.81 | 0.59 |
| Hamon | 0.89 | 1.50 | 0.72 | 0.17 | 1.31 |
| LSM–Hamon | 0.92 | 0.48 | 0.23 | 0.91 | 0.31 |
| GA–Hamon | 0.92 | 0.48 | 0.23 | 0.91 | 0.31 |
| PSO–Hamon | 0.91 | 1.21 | 0.57 | 0.38 | 1.04 |
| DE–Hamon | 0.92 | 0.48 | 0.23 | 0.91 | 0.31 |

Notes: HS, BR, Sch, Hamon are abbreviations for the Hargreaves–Samani model, the Baier–Robertson model, the Schendel model, and the Hamon model. LSM–HS, GA–HS, PSO–HS, and DE–HS are abbreviations for the Hargreaves–Samani model optimized by the least square method, the genetic algorithm, the particle swarm optimization algorithm, and the differential evolution algorithm; the same applies to BR, Sch, and Hamon. $R^2$, RMSE, RRMSE, NSE, and MAE are abbreviations for the determination coefficient, the root mean square error, the relative root mean square error, the Nash–Sutcliffe efficiency coefficient, and the mean absolute error.

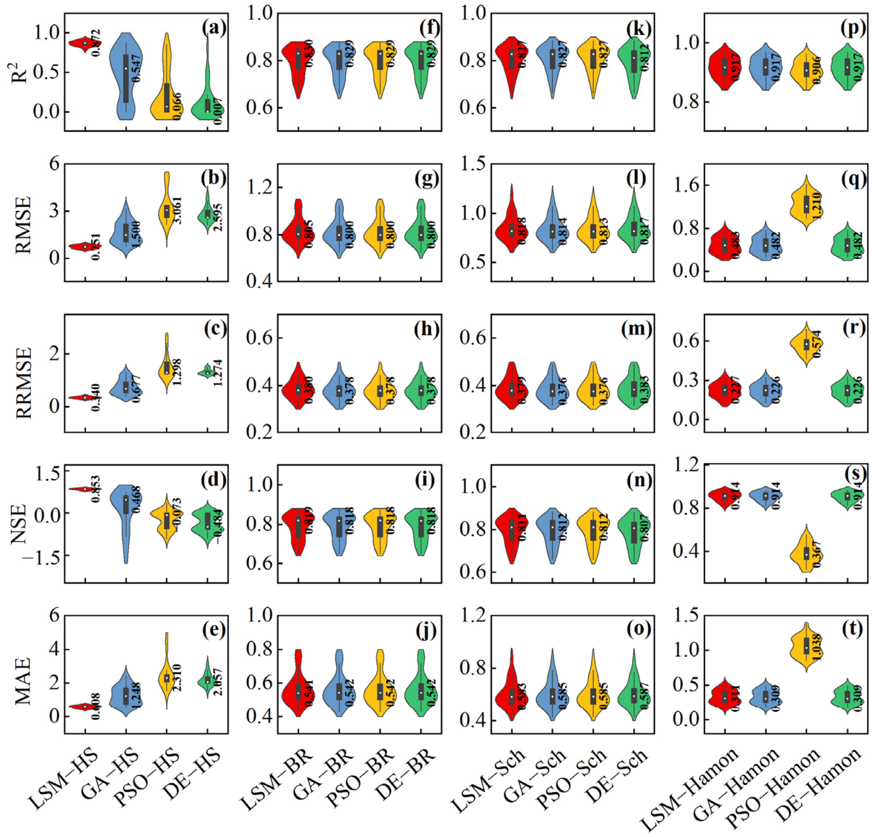

**Figure 5.** Violin plots of five evaluation metrics for temperature-based models. Notes: The boundary of the box represents the 25% and 75% values, which were the upper and lower interquartile (IQR), the error bar represents 1.5IQR, other values were considered outliers, and the dot represents the median. HS, BR, Sch, Hamon are abbreviations for the Hargreaves–Samani model, the Baier–Robertson model, the Schendel model, and the Hamon model. LSM–HS, GA–HS, PSO–HS, and DE–HS are the abbreviations for the Hargreaves–Samani model optimized by the least square method, the genetic algorithm, the particle swarm optimization algorithm, and the differential evolution algorithm; the same applies to BR, Sch, and Hamon. $R^2$, RMSE, RRMSE, NSE, and MAE are abbreviations for the determination coefficient, the root mean square error, the relative root mean square error, the Nash–Sutcliffe efficiency coefficient, and the mean absolute error.

Overall, while significantly enhancing the model's calculation accuracy, there was still optimization in the model's computation accuracy compared to LSM. Therefore, for the Sch model, both GA and PSO were suitable optimization algorithms, while the accuracy of the DE algorithm was relatively poorer. The Hamon model exhibited a significant enhancement in computational accuracy by LSM compared to the original model, with $R^2$, RMSE, RRMSE, NSE, and MAE being improved by 3.03%, 67.89%, 68.52%, 444.05%, and 76.28%, respectively. It was evident that LSM significantly enhances the computational accuracy of the Hamon model. In the Hamon model, the calculations using GA and DE yielded identical values for the five evaluation indicators. Furthermore, compared to LSM, $R^2$ and NSE were also identical to those of GA and DE, and RMSE and RRMSE differed from GA and DE by only 0.001. The MAE indexes of GA and DE were increased by 0.64% compared to LSM. Therefore, considering a comprehensive point of view, GA and DE were the most suitable algorithms for Hamon, while PSO exhibited the lowest computational accuracy among the four algorithms. In summary, the HS model was the best temperature-based model for Northeast China, and the Hamon model had higher simulation accuracy for ETo after GA optimization.

### 3.3. The Simulation Accuracy of Radiation-Based Models

Table 4 shows the medians of $R^2$, RMSE, RRMSE, NSE, and MAE for radiation-based models. The JH model had the highest calculation accuracy compared to the other three radiation-based models. The median values of $R^2$, RMSE, RRMSE, NSE, and MAE were superior to the PT and Mak models by 15.68%, 78.15%, 78.15%, 66.18%, and 82.89% and 19.45%, 84.61%, 83.73%, 164.24%, and 88.72%, respectively. It was better than the Door model by 6.64%, 98.03%, 98.02%, 782.52%, and 98.38% for $R^2$, RMSE, RRMSE, NSE, and MAE.

Figure 6 shows the violin plots for the $R^2$, RMSE, RRMSE, NSE, and MAE of ETo estimated by radiation-based models calibrated by LSM, GA, PSO, and DE. From Table 4 and Figure 6, it could be inferred that for the PT model, the computational accuracies of the original models were consistently lower than that of the LSM. The LSM enhanced the $R^2$ of the original model by 1.52%, enhanced the RMSE by 66.81%, enhanced the RRMSE by 67.17%, enhanced the NSE by 41.13%, and enhanced the MAE by 67.05%. It was evident that the LSM significantly improved the computational accuracy of the PT model. The $R^2$ and NSE of the original PT model were only lower than LSM and were higher than GA, PSO, and DE, whereas the RMSE, RRMSE, and MAE of the original PT model were the lowest. As the best-performing algorithm of the three optimization algorithms for the PT model, compared to LSM, PSO reduced $R^2$ by 18.93%, reduced RMSE by 53.38%, reduced RRMSE by 59.65%, reduced NSE by 27.07%, and reduced MAE by 54.79%. There was a significant disparity in computational accuracy compared to the LSM algorithm. Therefore, for the PT model, LSM has already achieved the goal of enhancing model accuracy, and the other optimization algorithms did not apply to the PT model. The Mak model exhibited the poorest performance for the other four computational accuracy evaluation indexes, except for $R^2$. This indicated that the Mak model did not have high computational accuracy, and algorithmic intervention was necessary to enhance its precision. LSM could significantly enhance the computational accuracy of the Mak model; for the original model, $R^2$ was improved by 4.96%, RMSE was improved by 76.69%, RRMSE was improved by 75.63%, NSE was improved by 125.29%, and MAE was improved by 78.42%. For the Mak model, the LSM had the greatest impact on enhancing the computational accuracy of the original model and was the most suitable algorithm for the Mak model. The GA, PSO, and DE did not enhance the computational accuracy of the Mak model as effectively as the LSM. The JH original model exhibited the highest computational accuracy among radiation-based models. The $R^2$ of the original model was equivalent to that of the LSM, while the NSE was better than the LSM by 0.002, and the MAE outperformed the LSM by 0.008. In contrast, the GA had the highest computational accuracy improvement for the JH model among the algorithms, improving the RMSE by 0.42%, the RRMSE by 0.46%, and the MAE by 0.95% compared with LSM. However, the $R^2$ was 0.001 lower than that of the LSM, and the NSE was 0.002 lower than the LSM and 0.003 lower than the original model. Overall, both the LSM and GA exhibited improvement over the original model within a certain range for the JH model. The computational accuracy of the two algorithms does not differ significantly, and both can be considered optimal algorithms for the JH model. For the Door model, the original model exhibited the lowest computational accuracy among radiation-based models, as well as among the various algorithms for calculating Door. The LSM improved the $R^2$, RMSE, RRMSE, NSE, and MAE by 6.76%, 98.08%, 98.07%, 784.47%, and 98.36% of the original model, respectively. It could be seen that LSM has greatly enhanced the computational accuracy of the original model. The GA, as the best-performing algorithm, improved RMSE, RRMSE, and MAE by 0.21%, 0.44%, and 1.28% compared to the LSM, while $R^2$ and NSE remained the same as the LSM. The GA could still further improve the model's accuracy despite the large improvement in the original model's computational accuracy achieved by the LSM. Therefore, the GA could be considered as the optimal algorithm for the Door model.

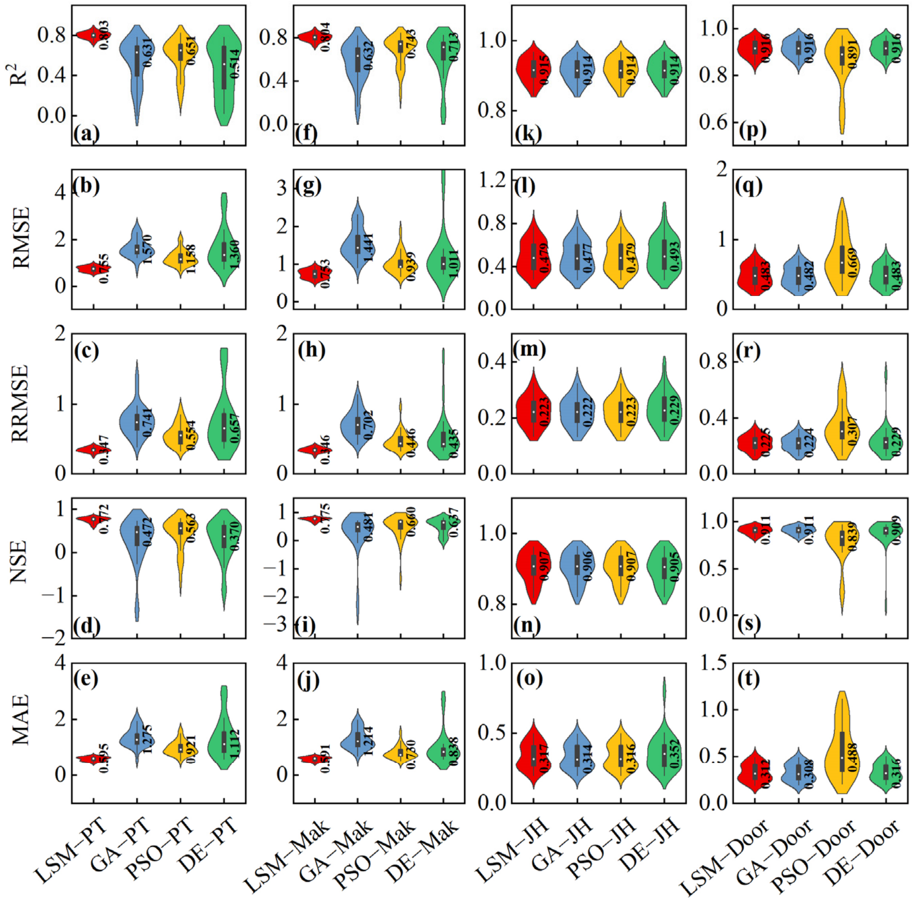


**Figure 6.** Violin plots of five evaluation metrics for radiation-based models. Notes: The boundary of the box represents the 25% and 75% values, which were the upper and lower interquartile (IQR), the error bar represents 1.5 IQR, other values were considered outliers, and the dot represents the median. PT, Mak, JH, and Door are abbreviations for the Priestley–Taylor model, the Makkink model, the Jensen and Haise model, and the Doorenbos and Pruitt model. LSM–PT, GA–PT, PSO–PT, and DE–PT are abbreviations for the Priestley–Taylor model optimized by the least square method, the genetic algorithm, the particle swarm optimization algorithm, and the differential evolution algorithm; the same applies to Mak, JH, and Door. $R^2$, RMSE, RRMSE, NSE, and MAE are abbreviations for the determination coefficient, the root mean square error, the relative root mean square error, the Nash–Sutcliffe efficiency coefficient, and the mean absolute error.

**Table 4.** The median values of the five evaluation criteria for radiation-based models.

| **Model** | **Model Accuracy Evaluation Metrics** | | | | |
|---|---|---|---|---|---|
| | $\mathbf{R^2}$ | **RMSE** | **RRMSE** | **NSE** | **MAE** |
| PT | 0.79 | 2.28 | 1.06 | 0.55 | 1.81 |
| LSM–PT | 0.80 | 0.76 | 0.35 | 0.77 | 0.60 |
| GA–PT | 0.63 | 1.57 | 0.74 | 0.47 | 0.65 |
| PSO–PT | 0.65 | 1.16 | 0.55 | 0.56 | 0.92 |
| DE–PT | 0.51 | 1.36 | 0.66 | 0.37 | 1.11 |

**Table 4.** *Cont.*

| Model | Model Accuracy Evaluation Metrics | | | | |
|---|---|---|---|---|---|
| | $R^2$ | RMSE | RRMSE | NSE | MAE |
| Mak | 0.77 | 3.23 | 1.42 | 0.34 | 2.74 |
| LSM–Mak | 0.80 | 0.75 | 0.35 | 0.78 | 0.59 |
| GA–Mak | 0.63 | 1.44 | 0.70 | 0.48 | 1.21 |
| PSO–Mak | 0.74 | 0.94 | 0.45 | 0.66 | 0.73 |
| DE–Mak | 0.71 | 1.01 | 0.44 | 0.64 | 0.84 |
| JH | 0.92 | 0.50 | 0.23 | 0.91 | 0.31 |
| LSM–JH | 0.92 | 0.50 | 0.22 | 0.91 | 0.32 |
| GA–JH | 0.92 | 0.48 | 0.22 | 0.91 | 0.31 |
| PSO–JH | 0.91 | 0.48 | 0.22 | 0.91 | 0.32 |
| DE–JH | 0.91 | 0.49 | 0.23 | 0.91 | 0.35 |
| Door | 0.86 | 25.2 | 11.7 | 0.10 | 19.0 |
| LSM–Door | 0.92 | 0.48 | 0.23 | 0.91 | 0.31 |
| GA–Door | 0.92 | 0.48 | 0.22 | 0.91 | 0.31 |
| PSO–Door | 0.89 | 0.67 | 0.31 | 0.84 | 0.49 |
| DE–Door | 0.92 | 0.48 | 0.23 | 0.91 | 0.32 |

Notes: PT, Mak, JH, and Door are the abbreviations for the Priestley–Taylor model, the Makkink model, the Jensen and Haise model, and the Doorenbos and Pruitt model. LSM–PT, GA–PT, PSO–PT, and DE–PT are abbreviations for the Priestley–Taylor model optimized by the least square method, the genetic algorithm, the particle swarm optimization algorithm, and the differential evolution algorithm; the same applies to Mak, JH, and Door. $R^2$, RMSE, RRMSE, NSE, and MAE are abbreviations for the determination coefficient, the root mean square error, the relative root mean square error, the Nash–Sutcliffe efficiency coefficient, and the mean absolute error.

On the whole, radiation-based models showed that the LSM consistently enhanced the computational accuracy of the PT, Mak, and Door models. For the PT and Mak models, the LSM stood as the optimal algorithm, while the optimal algorithm for the Door model was the GA. For the JH model, the calculation accuracy of the original model was almost the same as that of the LSM. The computational accuracy has already reached a high level, and the optimization algorithm did not significantly improve the accuracy.

### *3.4. The Simulation Accuracy of Combination Models*

Table 5 shows the median values of the five evaluation criteria—$R^2$, RMSE, RRMSE, NSE, and MAE—for combination models. The BS model had the best calculation accuracy in combination models, and the medians of $R^2$, RMSE, RRMSE, NSE, and MAE were superior to 48PM and PVB models by 0.80%, 97.58%, 97.53%, 1094.52%, and 97.72% and 19.65%, 72.67%, 73.50%, 252.45%, and 75.49%, respectively.

Figure 7 shows violin plots of the computational accuracy evaluation indexes from the four algorithms of LSM, GA, PSO, and DE for the combination models. By considering Table 5 and Figure 7, it could be inferred that for 48PM, the computational accuracy of the original models was lower than LSM. The LSM significantly improved the $R^2$, RMSE, RRMSE, NSE, and MAE of the original models by 0.68%, 97.88%, 97.76%, 1089.04%, and 98.06%, respectively. The LSM greatly enhanced the computational accuracy of the original models. The $R^2$ of the original model was only lower than that of the LSM and higher than that of the GA, PSO, and DE. However, the RMSE, RRMSE, NSE, and MAE were all lower than those of the three optimization algorithms. It is clear from Figure 7 that there was a substantial variance in computational precision among the algorithms. Overall, the computational accuracy of the 48PM original model was not high, and it could be greatly improved through algorithms. Among them, the LSM was the optimal algorithm, and PSO had the lowest computational accuracy among the four algorithms. For the PVB model, the original model exhibited the lowest computational accuracy. The LSM improved the original model's $R^2$ by 12.06%, its RMSE by 66.23%, its RRMSE by 66.98%, its NSE by

242.83%, and its MAE by 71.62%. The LSM significantly improved the computational accuracy of the PVB model. Both the GA and PSO further improved the computational accuracy of the model based on the LSM. The PSO increased the RMSE, RRMSE, and MAE of the LSM by 1.34%, 2.08%, and 0.73%, respectively, the $R^2$ was the same as that of the LSM, and the NSE was only 0.001 lower than that of the LSM. The $R^2$, NSE, and MAE of the GA and PSO were the same, and the difference between the RMSE and RRMSE was 0.001. This indicated that the GA and PSO had almost identical computational accuracy, and both could be used as the optimal algorithm for PVB. For the BS model, all model accuracy evaluation indexes, except for $R^2$, were worse than those of the original model. The $R^2$ of the LSM was the same as that of the original model, the RMSE was improved by 6.80%, the RRMSE was improved by 2.59%, the NSE was improved by 0.80%, and the MAE was improved by 6.53%. The LSM exhibited a significant improvement in computational accuracy compared to the original model. From Figure 7, it could be observed that the GA, PSO, and DE differ from the LSM by no more than 0.004 in terms of $R^2$, RMSE, RRMSE, NSE, and MAE. Therefore, it was evident that the computational accuracy of the three optimization algorithms was almost identical to the LSM, and all of them could be used as the optimal algorithms of the BS model to improve the computational accuracy of the original model. In summary, the BS model was the best combination model for Northeast China. The BS and 48PM models had higher simulation accuracy for ETo after LSM optimization.

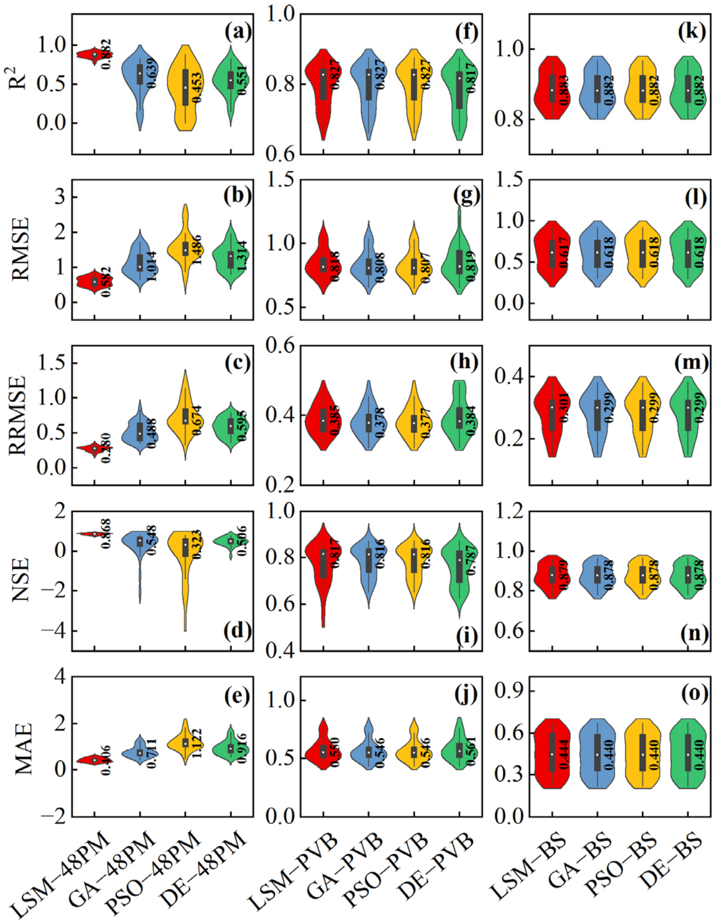


**Figure 7.** Violin plots of five evaluation metrics for combination models. Notes: The boundary of the box represents the 25% and 75% values, which were the upper and lower interquartile (IQR), the error bar represents 1.5IQR, other values were considered outliers, and the dot represents the median. 48PM, PVB, and BS are abbreviations for the Penman (1948) model, the Penman model improved by Van Bavel, and the Brutsaert and Stricker model. LSM–48PM, GA–48PM, PSO–48PM, and DE–48PM

are abbreviations for the Penman (1948) model optimized by the least square method, the genetic algorithm, the particle swarm optimization algorithm, and the differential evolution algorithm; the same applies to PVB and BS. $R^2$, RMSE, RRMSE, NSE, and MAE are abbreviations for the determination coefficient, the root mean square error, the relative root mean square error, the Nash–Sutcliffe efficiency coefficient, and the mean absolute error.

**Table 5.** The median values of the five evaluation indicators for combination models.

| Model | Model Accuracy Evaluation Metrics | | | | |
|---|---|---|---|---|---|
| | $R^2$ | RMSE | RRMSE | NSE | MAE |
| 48PM | 0.88 | 27.4 | 12.5 | 0.07 | 20.9 |
| LSM–48PM | 0.88 | 0.58 | 0.28 | 0.87 | 0.41 |
| GA–48PM | 0.64 | 1.01 | 0.49 | 0.55 | 0.71 |
| PSO–48PM | 0.45 | 1.49 | 0.67 | 0.32 | 1.12 |
| DE–48PM | 0.55 | 1.31 | 0.60 | 0.51 | 0.92 |
| PVB | 0.74 | 2.42 | 1.17 | −0.57 | 1.94 |
| LSM–PVB | 0.83 | 0.82 | 0.39 | 0.82 | 0.55 |
| GA–PVB | 0.83 | 0.81 | 0.38 | 0.82 | 0.55 |
| PSO–PVB | 0.83 | 0.81 | 0.38 | 0.82 | 0.55 |
| DE–PVB | 0.82 | 0.82 | 0.38 | 0.79 | 0.56 |
| BS | 0.88 | 0.66 | 0.31 | 0.87 | 0.48 |
| LSM–BS | 0.88 | 0.62 | 0.30 | 0.88 | 0.44 |
| GA–BS | 0.88 | 0.62 | 0.30 | 0.88 | 0.44 |
| PSO–BS | 0.88 | 0.62 | 0.30 | 0.88 | 0.44 |
| DE–BS | 0.88 | 0.62 | 0.30 | 0.88 | 0.44 |

Notes: 48PM, PVB, and BS are abbreviations for the Penman (1948) model, the Penman model improved by Van Bavel, and the Brutsaert and Stricker model. LSM–48PM, GA–48PM, PSO–48PM, and DE–48PM are abbreviations for the Penman (1948) model optimized by the least square method, the genetic algorithm, the particle swarm optimization algorithm, and the differential evolution algorithm; the same applies to PVB and BS. $R^2$, RMSE, RRMSE, NSE, and MAE are abbreviations for the determination coefficient, the root mean square error, the relative root mean square error, the Nash–Sutcliffe efficiency coefficient, and the mean absolute error.

### *3.5. Comprehensive Comparison of Three Types of ETo Models*

The estimation of ETo using different types of ETo estimation formulas has its advantages and disadvantages. By employing algorithms, it was possible to enhance the calculation accuracy of different models while leveraging their respective strengths and guaranteeing the accuracy of the models. Therefore, it was necessary to explore the accuracy of various ETo estimation models under different algorithms and to conduct a cross-comparison of the three types of ETo estimation models.

Based on Tables 3–5, it was evident that before using the algorithms to optimize the model, the best computational accuracy of the original model was JH. The $R^2$ of the original model reached 0.92, the RMSE reached 0.50 mm $d^{-1}$, the RRMSE reached 0.23, the NSE reached 0.91, and the MAE reached 0.31 mm $d^{-1}$, which were significantly superior to the calculation accuracy of other models and even surpassed the calculation accuracy of other models after the application of optimization algorithms. Therefore, for the JH model, the implementation of the LSM or optimization algorithms does not significantly enhance the calculation accuracy of the original model. However, it could still be observed in Figure 6 that the GA algorithm could still improve the accuracy of the JH model to a certain extent, indicating that the use of optimization algorithms can continue to improve the accuracy of model estimation after the accuracy of the model has reached a certain level. In addition to the JH model, the subsequent models with good accuracy performance were the BS and HS models, which were the best-performing models in the combination and temperature-based models, respectively. The median values of BS for $R^2$, RMSE, RRMSE, NSE, and MAE reached 0.88, 0.66 mm $d^{-1}$, 0.31, 0.87, and 0.48 mm $d^{-1}$, respectively, whereas for HS, the median values for $R^2$, RMSE, RRMSE, NSE, and MAE reached 0.87, 0.86 mm $d^{-1}$, 0.39, 0.82, and 0.63 mm $d^{-1}$, respectively. For both BS and HS, the accuracy of the original model calculations was lower than that of the LSM. In the case of the BS model, except for $R^2$, the model calculations were lower than the algorithm. The difference in the median of each evaluation index among the four algorithms was not more than 0.005. For the BS model, the algorithm could significantly improve the accuracy of model calculations, but the differences between the algorithms were not significant. This feature was also reflected in the BR model. For the HS model, the accuracy of the LSM model was much higher than the rest of the optimization algorithms, and it was the most suitable algorithm for the HS model. In conclusion, in the case of high computational precision of the model itself, regardless of the type of model, the LSM could still improve the model's accuracy to a certain extent, with the improvement from optimization algorithms compared to the LSM not being particularly significant.

From all models, before the use of optimization algorithms, although temperature-based models did not have the highest computational accuracy among the original models, they had a very stable model computational accuracy, with all evaluation indexes ranking in the top 2/3 except for NSE. In the absence of algorithms, temperature-based models were more suitable for the estimation of ETo in Northeast China. After the use of optimization algorithms, it was found that the HS model in temperature-based models, the PT and Mak models in radiation-based models, and the 48PM model in combination models were better calculated using the LSM. Apart from these four models, the precision of model calculations could be enhanced by optimization algorithms. Among them, the GA was the most universal optimization algorithm, and most of the models could obtain higher model estimation accuracy by using the GA. Using the optimization algorithm, the accuracy of the Door and Hamon models could be greatly improved, with the median $R^2$ values increased to 0.92, respectively, the median RMSE values increased to 0.48 mm $d^{-1}$, the RRMSE values increased to 0.22 and 0.23, respectively, the NSE values increased to 0.91, respectively, and the MAE values increasing to 0.31 mm $d^{-1}$, respectively.

In summary, the Hamon, JH, and Door models optimized by a heuristic intelligence algorithm had better simulation accuracy. Among them, after the optimization of the GA and DE algorithms, the accuracy of the Door model was greatly promoted; the values of parameters $a_{Door}$ and $b_{Door}$ are 0.15–0.26 and 0.24–0.32 after GA optimization and 0.03–0.50 and 0.25–0.33 after DE optimization, respectively (Figure 8c). For the JH model, all three optimization algorithms could make the model achieve satisfactory accuracy; the value range of optimized parameters is shown in Figure 8b.

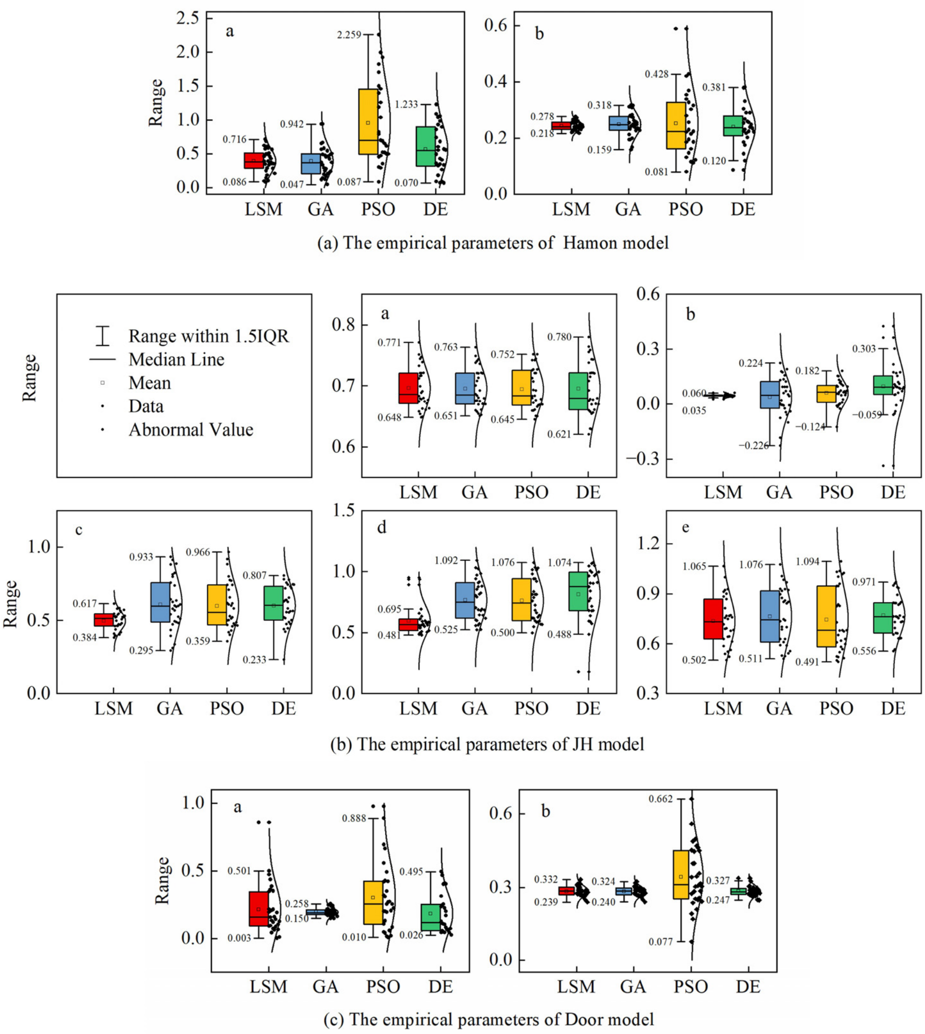

**Figure 8.** Parameters of recommended ETo estimation model obtained by different optimization algorithms in Northeast China. Notes: the boundary of the box represents the 25% and 75% values, which were the upper and lower interquartile (IQR), the error bar represents 1.5IQR, other values were considered outliers. The a, b, c, d, and e are empirical parameters.

## 4. Discussion

### *4.1. Reasons for Accuracy Discrepancies Among Different Models*

Correlation analysis showed that ETo was significantly correlated with $T_{mean}$, RH, and n in Northeast China, but the simulation accuracy of ETo empirical models based on different input variables was different in Northeast China. For temperature-based models, the HS model was one of the most direct empirical models for simulating ETo. Due to its high simulation accuracy and the fewer meteorological data variables required, the HS model was widely used worldwide [6]. In this study, the HS model exhibited superior accuracy compared to other models based on temperature; $R^2$ and NSE were 0.87 and 0.82, respectively. Bidabadi et al. [59] found that the use of $T_{min}$ and $T_{max}$, which are relatively easier to measure, responds better to the empirical Hargreaves model. Wu et al. [3] showed that the accuracy of the HS model in the temperate monsoon zone (TMZ) was better than that of the Sch and BR models, and the simulation results were the best among the temperature-based models. Xu and Singh [60] conducted a comparison of seven temperature-based ETo estimation models in Northwest Canada and concluded that the HS model could be recommended to estimate ETo in the region. The results of this study were similar to these.

The JH model demonstrated the most accurate estimation of ETo in Northeast China among all 11 models in this study. In the TMZ region, Wu et al. [61] showed that the PT model had superior accuracy compared to the JH model, while the result was the opposite in the temperate continental zone (TCZ). Both TMZ and TCZ include part of the region of Northeast China. The reason for the inconsistency between this study and the research results of Wu et al. may be the spatial difference of the study area. Feng et al. [25] showed that in the hilly region of central Sichuan, the ETo simulated by the PT model was nearly identical to that of the PM equation, and PM was minimal. However, the PT model was not the best empirical model for simulating ETo in Northeast China. This may be because the climate environment built by the PT model was a region with high air humidity, so it was more suitable for a region with a relatively humid climate, like Sichuan.

The combination model typically consists of a semi-empirical formula grounded in theory. This study showed that in Northeast China, the BS model had the best accuracy of the three combination models, which is consistent with the result of Wu et al. [61] in the TCZ region. Wu et al. [61] showed that for the Harbin site, the PT model showed greater estimation accuracy compared to the HS model, and the estimation accuracy of these two models was better than that of the Mak model. However, the results of this study showed that the calculation accuracy of the PT and Mak models was inferior to that of the HS models. This may be because their conclusions were based on a single site (Harbin), but this study was based on the entire Northeast China region. Xu and Singh [60] showed that due to the difference between the modeling conditions and the use conditions of the empirical model, the error of the model simulation results was large when the original empirical parameter values in the model were used for simulation. By calibrating the parameters using local data, the accuracy of the model could be significantly enhanced. The majority of studies found that the combination model had better simulation accuracy compared to the temperature- or radiation-based models. However, for Northeast China, the simulation accuracy based on the radiation model was better than that of other models, the simulation accuracy of the Hamon model based on temperature was also acceptable, but the accuracy of the combination model was the lowest. The reason may be that the combination models require the most input meteorological factors, resulting in large errors, which will increase the complexity of the model and reduce the accuracy of estimation. Considering that the simulation accuracy of models obtained by single-site and multi-site analysis at different locations, even in the same region, is not the same, the comparative analysis between different ETo estimation models in Northeast China is meaningful.

### *4.2. Reasons for Accuracy Discrepancies Among Different Intelligence Algorithms*

The results showed that the LSM can significantly promote the accuracy of temperature-based, radiation-based, and combination models. The LSM has good statistical properties, a solid mathematical theory foundation, and has found widespread application in numerous fields. The model optimized by the LSM can solve the problem of large differences in simulation accuracy in different regions. However, when there are many parameters, the LSM tends to have too many iterations, and it can easily become stuck in a locally optimal solution. Therefore, some optimization algorithms can be used in parameter fitting. In this study, three optimization algorithms—the GA, the PSO, and the DE—are selected to rate the model parameters. The input types of meteorological factors significantly affect the estimation accuracy of ETo. Jia et al. [62] found that machine learning models optimized by the PSO algorithm can input fewer meteorological factors and estimate ETo accurately, but under different climatic conditions, different meteorological factors have different effects on the accuracy of estimating ETo. This study showed that although the simulation accuracy of the HS and JH empirical models was lower than that of LSM optimization, its accuracy was higher than that of the models optimized by the three optimization algorithms. The HS and JH models need to input three or more parameters, and they are the models that need to input the most parameters in the temperature-based and radiation-based models, respectively. Too many input factors and parameters may lead to the introduction of more error terms in the calculation process of the algorithm, resulting in an increase in model complexity and overfitting in the process of optimization.

The calculation accuracy of other models was promoted after algorithm optimization, and the three models with the best accuracy were finally optimized as the temperature-based Hamon model and the radiation-based JH and Door models. The Hamon and Door models need fewer input factors, but the calculation accuracy of the original model was not high. However, the calculation accuracy of the original model could be improved to a high degree through the algorithm ($R^2$ about 0.92, RMSE about 0.48 mm $d^{-1}$, RRMSE about 0.23, NSE about 0.91, and MAE about 0.31 mm $d^{-1}$). Jalal Shiri et al. [63] showed that support vector machines based on the firefly algorithm have a high accuracy in estimating ETo in wet areas in temperature-based models. Due to global warming, the method of estimating ETo using temperature data alone introduces higher errors than before [64]. Song et al. [64] showed that temperature change promoted the increase of ETo by 59.5%, and if there was a method to reasonably evaluate the influence of temperature, it could provide a more accurate estimation of ETo. Therefore, it is necessary to consider the influence of climate warming on the calculation accuracy of the model when using the Hamon model in the future. The parameters of the JH model that need to be calibrated are larger than those of the Hamon and Door models, but its simulation accuracy is the highest among all 11 empirical models. The optimization algorithm promoted the calculation accuracy of the JH model slightly, and it could make the calculation accuracy of the JH model more stable.

### *4.3. Analysis of Uncertainties, Limitations, and Applications*

In this study, only several widely used ETo estimation empirical models were selected, and these models were divided into temperature-based, radiation-based, and combination models according to the required input meteorological factors. There are more types of ETo estimation models than these three, including models based on humidity, aerodynamics, and mass transfer [23,65–67]. The results of this study showed that the radiation-based model had a better simulation effect for ETo estimation in Northeast China, but this conclusion was only based on the current investigation results. There may be ETo estimation models with higher simulation accuracy that are more suitable for future exploration in Northeast China.

In addition, for radiation-based models, this study only used optimization algorithms to perform parameter calibration on the empirical model estimating ETo to enhance the accuracy of the model. However, there are various calculation methods for radiation as the input variable of the radiation-based models [68]. The accuracy of different formulas may vary significantly in different regions. The Angstrom–Prescott formula based on sunshine duration [69] is the standard method recommended by FAO 56 [6]. In this study, only the standard method was used to calculate the radiation, and the accuracy difference between different radiation calculation methods was not taken into account. In future research, the accuracy of radiation calculation can be further improved by the algorithm under the condition that the accuracy of the radiation-based ETo estimation model was known to be the highest, and an ETo estimation model with higher accuracy can be obtained in Northeast China.

Despite the above uncertainties and limitations of this study, the results provide an important theoretical reference for the management of agricultural water resources in Northeast China and a valuable resource for professionals. First of all, this study analyzed the temporal and spatial changes of meteorological factors and ETo in Northeast China in the past 60 years and more accurately understood the climate characteristics and hydrological conditions in Northeast China, which helps optimize the irrigation scheme in the region, adjust the planting structure of crops, and improve the utilization efficiency and production level of cultivated land. Secondly, accurate estimation of ETo can help us develop more reasonable irrigation plans, ensure accurate irrigation of crops in the growing season, improve crop yield and quality, reduce water waste and soil salinization risk, and improve the sustainability and competitiveness of production in Northeast China. In addition, the research results can also provide scientific support for the construction of farmland water conservancy facilities and the formulation of water resources management policies in Northeast China. Through in-depth study of ETo change rules and estimation methods, we can better predict the impact of climate change on agricultural water resources, propose corresponding countermeasures, strengthen water resources management and protection, and promote the sustainable use and management of agricultural water resources. In short, this study has important significance and value for the practical application of agricultural water management in Northeast China and can provide a scientific basis and technical support for improving agricultural production efficiency, protecting water resources and the environment, and promoting the sustainable development of agriculture.

In future research and application, we can further expand the research scope and try more models and algorithms to optimize the simulation accuracy of variables such as radiation in the model to improve the reliability of ETo estimation and reduce the limitations of this study. In doing so, future studies would closely integrate with agricultural water management practices and needs, strengthen communication and cooperation with the actual needs of agricultural production, promote the application and promotion of research results, provide better scientific support and decision-making references for agricultural water management in different regions, and promote the sustainable development of agriculture and the effective use of water resources.

## 5. Conclusions

As one of China's crucial grain production bases, Northeast China's agricultural water management plays a vital role in national food security. This study evaluated and optimized various ETo estimation models to improve agricultural water management efficiency in the region. Through comprehensive analysis and comparison, our findings revealed distinct patterns of model performance across different categories. Among the empirical models evaluated, the Hamon and Hargreaves–Samani models demonstrated superior performance in temperature-based approaches, while the Jensen–Haise and Door models exhibited exceptional estimation capabilities in radiation-based methods. In the combination model category, the Brutsaert–Stricker model showed the most promising results. Overall, the Jensen–Haise model emerged as the most reliable among all tested empirical models for Northeast China's conditions.

The application of optimization algorithms, particularly the GA and DE, significantly enhanced model performance. The Hamon and Door models showed remarkable improvement after optimization, demonstrating the potential of these algorithms in refining ETo estimation. The Jensen–Haise model maintained consistently high performance across all optimization algorithms, indicating its robust nature and reliability for regional applications. Based on these findings, the GA-optimized Hamon model is recommended when only temperature data are available, while the GA-optimized Door model proves most effective when only radiation data can be accessed. In cases where comprehensive meteorological data are available, the algorithm-optimized Jensen–Haise model presents the most dependable solution for ETo estimation in Northeast China.

This study focused primarily on model simulation accuracy; however, the precision of input factor calculations, such as $R_s$, significantly influences model performance. Future research should explore optimal model–algorithm combinations across different regions.

**Author Contributions:** Z.L.: writing—original draft, conceptualization, methodology, software. Z.Z.: writing—review and editing. L.X.: editing and funding acquisition, project administration. L.Z.: data curation, investigation. N.C.: data curation, investigation. H.C.: editing and funding acquisition, project administration. All authors have read and agreed to the published version of the manuscript.

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## Nomenclature

| | | | |
|---|---|---|---|
| Abbreviations | | TMZ | Temperate monsoon zone |
| ANN | Artificial Neural Networks | 48PM | Penman model (1948) |
| BR | Baier–Robertso model | Variables | |
| BS | Brutsaert and Stricker model | $e_a$ | Actual water vapor pressure (kPa) |
| DE | Differential evolution algorithm | $e_s$ | Saturation water vapor pressure (kPa) |
| ELM | Extreme learning machine | G | Soil heat flux (MJ $m^{-2}$ $day^{-1}$) |
| ET | Evapotranspiration | N | Maximum possible sunshine hours (h) |
| ETo | Reference crop evapotranspiration | n | Sunshine duration (h) |
| FAO | The World Food and Agriculture Organization | P | Precipitation (mm) |
| GA | Genetic algorithm | $P_g$ | Global optimal position of all particles |
| HS | Hargreaves–Samani model | $P_i$ | Individual optimal position of particle i |
| JH | Jensen and Haise model | $R_a$ | Extraterrestrial radiation (MJ $m^{-2}$ $day^{-1}$) |
| LSM | Least square method | RH | Relative humidity (%) |
| MAE | Mean absolute error | $R_n$ | Net radiation (MJ $m^{-2}$ $day^{-1}$) |
| Mak | Makkink model | $R_s$ | Solar shortwave irradiance (MJ $m^{-2}$ $day^{-1}$) |
| NMIC | The National Meteorological Information Center | $T_{max}$ | Maximum temperature (°C) |
| NSE | Nash–Sutcliffe efficiency coefficient | $T_{mean}$ | The daily mean temperature (°C) |
| PM | Penman–Monteith method | $T_{min}$ | Minimum temperature (°C) |
| PSO | Particle swarm optimization algorithm | $u_{10}/u_2$ | Wind speed at 10/2 m height (m $s^{-1}$) |
| PT | Priestley–Taylor model | $v_i(t)$ | Velocity of particle i at the t–th iteration |
| PVB | Penman model improved by Van Bavel | $X_i(t)$ | Position of particle i at the t–th iteration |
| R | Ritchie model | $\gamma$ | Psychrometer constant (kPa $°C^{-1}$) |
| RF | Random Forests | $\lambda$ | Latent heat of evaporation (MJ $kg^{-1}$) |
| RMSE | Root mean square error | $\omega$ | Inertia weight |
| RRMSE | Relative root mean square error | $\Delta$ | Inclination rate of saturated water vapor pressure curve (kPa $°C^{-1}$) |
| $R^2$ | Determination coefficient | Lat | Latitude (°) |
| Sch | Schendel model | Lon | Longitude (°) |
| TCZ | Temperate continental zone | H | Elevation (m) |

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